

PUNJAB RETROFITTING SCHOOLS AND HEALTH FACILITIES

Sector: Infrastructure & Urban Resilience

Hazard addressed: Flood

Targeted assets: Education facilities, Health facilities



Description: This measure involves structural retrofitting of public buildings – specifically schools and colleges – located near river banks across 9 Punjab districts. Interventions include improving site drainage and constructing flood barriers around qualifying buildings.

Priority Districts: Lahore, Sheikhupura, Nankana Sahib, Jhang, Toba Tek Singh, Multan, Muzaffargarh, Bahawalpur, Rajanpur

TECHNICAL SPECIFICATIONS/DIMENSION



Unit of implementation: Per building.

Scale assumptions: Educational buildings (schools and colleges) within 250m of river lines in target districts.

Key design parameters: Improving site drainage; construction of flood barriers; eligibility criterion = within 250m of river lines.

Time horizon: 2026–2040

IMPACT



Risk reduction level: High Effectiveness – significant reduction (Expert Rank 2). Damages are reduced by 90% (MDD=0.1) corresponding to 1-in-100-year flood events.

Time to effectiveness: Medium-term; effectiveness realized progressively as individual buildings are strengthened over three implementation phases (2026–2040).

Expected impact: Expected to significantly reduce structural flood damage to educational buildings located in close proximity of rivers through flood barriers and drainage improvement.

IMPLEMENTATION CONSIDERATIONS



Enabling conditions: Structural engineering assessments for each building; hydraulic design for flood barriers; asset surveys to identify eligible buildings; inter-departmental coordination between Education and Irrigation departments.

Required institutions: Punjab School Education Department; Punjab Higher Education Department; Punjab Irrigation Department (for river proximity mapping); PDMA Punjab.

Barriers: Asset survey and eligibility mapping complexity; procurement of specialized contractors; coordination across multiple departments.

COST ASSUMPTIONS



Unit cost: Not a single unit cost used, cost varies by building size.

Initial Investment Cost (CAPEX):
USD 83,745,598.

Annual O&M costs (OPEX):
USD 133 per km per year.

CO-BENEFITS & TRADE-OFFS



Environmental: Improved site drainage at school/college campuses may reduce localized waterlogging; flood barriers can help protect riparian green spaces.

Social: Protected school and college infrastructure ensures continuity of education services during flood events; reduces displacement and trauma for students; schools often serve as temporary flood shelters.

Economic: Avoided costs of flood damage and reconstruction of buildings; continuity of education services reduces longer-term human capital losses.

Potential risks: CAPEX assumes existing structures can be adequately protected; some buildings may require full reconstruction and/ or relocation rather than retrofitting.

BENEFICIARIES



Primary beneficiaries: Students and staff in schools and colleges; communities using these facilities as flood shelters; Punjab Education departments.

Distributional aspects: Benefits accrue disproportionately to communities living near rivers.